

CSCI-473: Applied Data Science

Homework Assignment 1

Cleaning and Summarizing Palmer Penguins Data

Weight:	5% of the final course grade
Assignment Type:	Individual
Programming Language:	Python
Due Date:	Monday, October 5, 2026, at 11:59 PM EST
Instructor:	Desta Haileselassie Hagos, PhD
Questions:	desta.hagos@howard.edu

Important

Read the complete assignment instructions before beginning your work. Submit exactly **one Python source file** and **one PDF report** through Canvas. This is an individual assignment.

1 Purpose of the Assignment

This homework gives you practice applying the data-science workflow to a real dataset. You will use pandas to inspect, clean, validate, summarize, and export data. The assignment focuses on concepts covered in Lectures 1–4, including:

- loading CSV data with `pd.read_csv()`;
- inspecting DataFrame structure, data types, and summary statistics;
- selecting, filtering, and sorting rows and columns;
- identifying and handling missing values;
- cleaning column names and categorical values;
- converting data types;
- checking duplicates and validating values;
- using `groupby()` and `agg()`; and
- saving a cleaned dataset with `to_csv()`.

You are not required to build a machine-learning model or create visualizations in this assignment.

2 Dataset

You will work with the **Palmer Penguins raw dataset**. The data were collected by Dr. Kristen Gorman and the Palmer Station, Antarctica Long-Term Ecological Research program. The raw dataset contains measurements and observational information for **344 penguins** representing three species and three islands. The dataset includes:

- numerical measurements such as culmen length, culmen depth, flipper length, and body mass;
- categorical variables such as species, island, sex, and clutch completion;
- a date column;
- identifiers; and
- missing values that require inspection.

The dataset is distributed under a CC0 license. Additional information is available from the official Palmer Penguins documentation:

<https://allisonhorst.github.io/palmerpenguins/>

Dataset File

Download `palmer_penguins_raw.csv` from the Homework 1 assignment page on Canvas. Save the CSV file in the same folder as your Python file. You should use the version provided on Canvas so that everyone works with the same dataset.

3 General Programming Requirements

Your program must:

- use pandas;
- complete all six tasks in the order presented;
- contain clear comments identifying each task;
- use descriptive variable names;
- run from beginning to end without errors;
- print clearly labeled results for every task; and
- save the final cleaned dataset as `palmer_penguins_clean.csv`.

Do not manually calculate results that pandas can produce. Your reported answers must be generated by your Python program.

4 Task 1: Load and Audit the Dataset

Load the provided CSV file into a pandas DataFrame named `df`.

Your program must display:

1. the first five rows;
2. the last five rows;
3. the number of rows and columns;

4. all column names;
5. the data type of every column;
6. the output of `info()`;
7. descriptive statistics for the numerical columns; and
8. the number of missing values in each column.

After inspecting the output, identify at least **three data-quality issues or concerns** that should be investigated. A concern does not have to be an error. For example, an unexpectedly formatted category or a column with many missing values may require investigation.

Code Hint

```
import pandas as pd

df = pd.read_csv("palmer_penguins_raw.csv")

print("Shape:", df.shape)
print("\nMissing values:")
print(df.isna().sum())
```

Report Requirement

Explain what the audit revealed. Include at least one readable screenshot showing the structural inspection and missing-value counts.

5 Task 2: Clean Column Names and Correct Data Types

Create a working copy of the original DataFrame before cleaning:

```
raw_df = df.copy()
clean_df = df.copy()
```

Rename the columns so that they use clear `snake_case` names. Your final names should include the following:

Original Name	Required Clean Name
studyName	study_name
Sample Number	sample_number
Species	species
Island	island
Clutch Completion	clutch_completion
Date Egg	date_egg
Individual ID	individual_id
Culmen Length (mm)	culmen_length_mm
Culmen Depth (mm)	culmen_depth_mm
Flipper Length (mm)	flipper_length_mm
Body Mass (g)	body_mass_g
Sex	sex

You may also rename the remaining columns using the same style. Then:

1. convert `date_egg` to a pandas date-time data type. Use `pd.to_datetime()` and set conversion errors to `"coerce"`;
2. use `pd.to_numeric()` with `errors="coerce"` on the four primary measurement columns to ensure that they are numerical; and
3. print the data types after conversion.

The four primary measurement columns are:

- `culmen_length_mm`,
- `culmen_depth_mm`,
- `flipper_length_mm`, and
- `body_mass_g`.

Code Hint

```
clean_df = clean_df.rename(columns={
    "studyName": "study_name",
    "Sample Number": "sample_number",
    "Date Egg": "date_egg"
    # Continue the dictionary for the remaining columns.
})

clean_df["date_egg"] = pd.to_datetime(
    clean_df["date_egg"],
    errors="coerce"
)
```

Report Requirement

Describe the name and type corrections you made. Include a screenshot showing the cleaned column names and resulting data types.

6 Task 3: Clean Categories, Missing Values, and Duplicates

Complete the following cleaning steps.

A. Inspect and standardize categories

Use `unique()` or `value_counts(dropna=False)` to inspect:

- `species`,
- `island`,
- `clutch_completion`, and
- `sex`.

Replace any placeholder value in `sex`, such as `"."`, with `pd.NA`. Standardize the species values to the shorter category names:

- `Adelie`,
- `Chinstrap`, and
- `Gentoo`.

Do not assume that categories are consistent. Inspect them before applying a mapping or replacement.

B. Check duplicates

Report:

1. the number of exact duplicate rows; and
2. all occurrences of repeated `individual_id` values. Use `duplicated()` with the `subset` argument set to the identifier column and `keep=False`.

Explain why a repeated identifier is not automatically the same as an exact duplicate row. Remove exact duplicates if any are present. Do not remove rows only because `individual_id` appears more than once unless you can justify that decision from the complete row.

C. Handle missing values

For the final analysis dataset, keep only rows that contain recorded values for all of the following required fields:

- `species`,
- `island`,
- `culmen_length_mm`,
- `culmen_depth_mm`,
- `flipper_length_mm`,

- `body_mass_g`, and
- `sex`.

Use `dropna(subset=...)` rather than removing every row that contains any missing value anywhere in the dataset.

Report the number of rows before and after this step. Briefly explain why `subset` is important here.

Code Hint

```
required_columns = [  
    "species",  
    "island",  
    "culmen_length_mm",  
    "culmen_depth_mm",  
    "flipper_length_mm",  
    "body_mass_g",  
    "sex"  
]  
  
clean_df = clean_df.dropna(  
    subset=required_columns  
) .copy()
```

Report Requirement

Explain how you standardized categories, evaluated duplicates, and handled missing values. Include at least one screenshot showing the category counts, duplicate checks, and the number of rows before and after cleaning.

7 Task 4: Validate the Cleaned Data

Perform the following validation checks:

1. verify that the four primary measurement columns contain only positive values;
2. verify that `species` contains only `Adelie`, `Chinstrap`, and `Gentoo`;
3. verify that `island` contains only the categories observed in the original dataset;
4. verify that the required fields from Task 3 contain no missing values; and
5. verify that the cleaned dataset contains no exact duplicate rows.

Use Boolean conditions and at least **three assertions**. Do not invent biological upper limits for measurements unless a documented rule is provided. A large value is not automatically invalid.

Code Hint

```
allowed_species = {"Adelie", "Chinstrap", "Gentoo"}

assert set(
    clean_df["species"].unique()
).issubset(allowed_species)

assert (clean_df["body_mass_g"] > 0).all()
assert clean_df[required_columns].notna().all().all()
assert clean_df.duplicated().sum() == 0
```

Print a confirmation message after all assertions pass.

Report Requirement

List the rules you validated and explain what each assertion checks. Include a screenshot showing the successful validation output.

8 Task 5: Filter, Sort, and Summarize the Data

Use the cleaned DataFrame to answer all of the following questions.

A. Filtering and sorting

1. Display all Gentoo penguins with a body mass greater than 5,000 grams.
2. Display all female Adelie penguins from Torgersen Island.
3. Display the five penguins with the greatest flipper lengths. Show the species, island, sex, flipper length, and body mass.

At least one filter must combine conditions using `&`. Remember to place each comparison inside parentheses.

B. Group summaries

Calculate and display:

1. the number of penguins in each species;
2. the mean body mass for each species;
3. the mean culmen length, mean flipper length, and mean body mass for each species; and
4. the number and mean body mass of penguins grouped by both species and sex.

Use `groupby()` and use `agg()` for the summaries requiring multiple statistics or columns. Round displayed means to two decimal places.

Code Hint

```
gentoo_heavy = clean_df[
    (clean_df["species"] == "Gentoo")
    & (clean_df["body_mass_g"] > 5000)
]

species_summary = clean_df.groupby("species").agg(
    penguin_count=("species", "size"),
    mean_body_mass_g=("body_mass_g", "mean")
)
```

The hints illustrate the required pandas operations. You must complete all questions and label every result clearly.

Report Requirement

For each question, state the pandas operation you used and briefly interpret the result. Include readable screenshots showing the filtering, sorting, and group-summary outputs.

9 Task 6: Export and Verify the Cleaned Dataset

Save the final cleaned DataFrame as:

`palmer_penguins_clean.csv`

Do not save the DataFrame index as a separate column. Reload the saved file into a new DataFrame and display:

- its first five rows;
- its shape; and
- its missing-value counts for the required fields.

Compare the reloaded shape with the final shape before saving. They should be the same.

Code Hint

```
clean_df.to_csv(  
    "palmer_penguins_clean.csv",  
    index=False  
)  
  
check_df = pd.read_csv(  
    "palmer_penguins_clean.csv"  
)
```

Report Requirement

Describe how you verified the exported file. Include a screenshot showing the reloaded data, its shape, and the missing-value check.

10 PDF Report Requirements

Submit one clearly organized PDF report. The report must contain the following sections.

1. Student Information

Include:

- full name;
- student ID number;
- Howard University email address;
- course: CSCI-473; and
- assignment title.

2. Introduction

Briefly describe the dataset and the purpose of the assignment.

3. Tasks 1–6

Create a separate, clearly labeled subsection for every task. For each task:

- describe what you did;
- identify the important pandas operations you used;
- report and briefly interpret the results;
- explain any cleaning decision you made; and
- include at least one readable screenshot of the relevant program output.

Screenshots must show the output produced by your submitted Python file. Screenshots of code alone do not satisfy this requirement. Crop screenshots so that the relevant output is readable.

4. Cleaning Summary

Summarize:

- the problems you found;
- how you handled them;
- how many rows were removed, if any; and
- how you verified the final dataset.

5. GenAI Use Statement

Include the statement described in the Responsible Use of Generative AI section of this document.

6. Conclusion

Briefly explain what you learned about inspecting, cleaning, validating, and summarizing real data.

Code in the Report

Do not paste your entire Python program into the PDF. The complete source code will be submitted separately. You may include short code excerpts when they help explain an important cleaning or validation decision.

11 Submission Requirements

Upload exactly the following two files directly to Canvas:

1. `Lastname.Firstname_HW1.py`
2. `Lastname.Firstname_HW1.Report.pdf`

The Python file must contain the complete solution for Tasks 1–6 and must run successfully on its own when the provided CSV file is in the same folder.

The cleaned CSV file is an output of your program and does **not** need to be uploaded unless the instructor requests it separately.

How to Submit

Upload the Python source file and PDF report directly to Canvas. Do not combine the files into a ZIP folder. Submissions sent by email will not be accepted or graded.

Before submitting, verify that:

- both filenames follow the required naming convention;
- the Python program runs from beginning to end without errors;

- all printed results are clearly labeled;
- every task is explained in the PDF report;
- all screenshots are readable; and
- the GenAI Use Statement is included.

12 Responsible Use of Generative AI

Responsible use of Generative AI tools is permitted for this assignment.

Examples of appropriate uses include:

- asking for clarification of a pandas concept;
- asking why an error occurred;
- receiving feedback on a cleaning approach;
- generating additional practice examples; and
- checking whether your explanation is clear.

You are responsible for the correctness of every line of code, every result, and every explanation that you submit.

Important

You must understand and be able to explain all submitted work. Do not submit AI-generated code or explanations that you have not independently checked. Do not use methods or libraries that have not been covered in class merely because a Generative AI tool suggested them.

If you used a Generative AI tool, your PDF report must identify:

- the tool you used;
- what you used it for;
- one or more representative prompts or questions you submitted;
- how you checked the accuracy of the response; and
- what you changed or corrected, when applicable.

If you did not use a Generative AI tool, state:

I did not use a Generative AI tool while completing this assignment.

13 Academic Integrity

This is an individual assignment. You may discuss general course concepts with classmates, but the submitted code, analysis, explanations, and report must be your own work.

You may not:

- copy another student's code, report, screenshots, or results;
- share your completed solution with another student;
- divide the tasks among students;
- copy a complete solution from an online repository;
- fabricate output or screenshots; or
- submit AI-generated work that you do not understand.

You should be prepared to explain any part of your submitted program or report if asked by the instructor.

14 Grading

Homework Assignment 1 is worth 5% of the final grade of the course and will be graded out of 5 points.

Component	Points
Task 1: Loading and initial audit	0.50
Task 2: Column names and data types	0.50
Task 3: Categories, missing values, and duplicates	1.00
Task 4: Validation and assertions	0.50
Task 5: Filtering, sorting, grouping, and interpretation	1.00
Task 6: Exporting and verifying the cleaned dataset	0.50
PDF report, screenshots, organization, and clarity	0.50
GenAI Use Statement and submission requirements	0.50
Total	5.00

Partial credit may be awarded when the submitted work demonstrates a correct approach but contains a limited implementation or interpretation error.

15 Final Checklist

- ☐ I used the dataset provided through Canvas.
- ☐ My single Python file completes Tasks 1–6.
- ☐ My program runs without errors.
- ☐ My outputs are clearly labeled.
- ☐ My PDF explains every task and includes readable screenshots.

- ☐ I included a GenAI Use Statement.
- ☐ I used the required filenames.
- ☐ I uploaded one .py file and one PDF file to Canvas.